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RESEARCH

Dal-yolo: a multi-target detection model for UAV-based road maintenance integrating feature pyramid and attention mechanisms

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Abstract
Routine road maintenance is essential to ensure traffic safety. This study proposes DAL-YOLO, a novel multi-object detection model specifically designed for UAV-based road maintenance. First, the feature extraction module is enhanced markedly in small object detection accuracy by integrating Deformable Attention into the C3k2 module. Second, an Adaptive Scaled Pyramid Network is introduced, dynamically fusing high-, mid-, and low-level features with adaptive fusion weights, excelling in dense targets and complex scenarios. Furthermore, a lightweight detection head, the Lightweight Shared Convolution BatchNorm Head, combines shared convolution layers with independent BatchNorm structures, improving precision and recall for dense and occluded object detection substantially. Experimental results on the road maintenance dataset show that DAL-YOLO achieves an AP of 39.8% and an AP50 of 58.8%, improving the accuracy of small, medium, and large objects notably. This model offers an efficient and accurate solution for UAV-based routine road maintenance tasks.

Keywords UAV · Multi-object detection · Road maintenance · Multi-scale feature fusion · Dense target detection

1 Introduction

Routine road maintenance is essential for ensuring public road safety and promoting sustainable development. As urbanization rapidly advances and traffic volume continues to grow, road maintenance tasks have become more demanding [1]. Traditional manual inspection methods require significant labor and time, are prone to subjective biases, and fail to meet modern requirements for efficiency, accuracy, and diversity in road maintenance tasks [18].

In recent years, advancements in artificial intelligence (AI) and computer vision technologies have provided innovative solutions for intelligent road maintenance [15]. Among these, unmanned aerial vehicles (UAVs) have gained attention due to their efficiency, flexibility, and cost-effectiveness. UAVs have demonstrated promising results in single-object detection scenarios such as pavement defect identification, traffic sign recognition, and road debris detection [3]. However, real-world road maintenance tasks involve detecting multiple targets, including pavement defects, infrastructure damage, and safety hazards. Single-object detection methods are inadequate for such scenarios. Moreover, UAV-captured road images often feature targets with diverse scales, complex backgrounds, and significant variations in characteristics, posing substantial challenges to existing multi-object detection algorithms.

Current object detection algorithms are broadly categorized into two-stage and single-stage detection methods. Two-stage methods, such as Faster R-CNN [17] and Mask R-CNN [6], are renowned for their high accuracy but are computationally intensive, limiting their suitability for real-time applications. In contrast, single-stage methods, such as Single Shot Multibox Detector (SSD) [13], RetinaNet [12], and the YOLO series, are more efficient and better suited for multi-object detection tasks in UAV scenarios. YOLO11 [9], the latest iteration of the YOLO series, exhibits significant improvements in model architecture, feature extraction, and multi-scale object detection. However, YOLO11, while excelling in generalized tasks, struggles with detecting multiple objects of varying scales in complex backgrounds, a limitation observed in many existing multi-object detection

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algorithms. Consequently, existing methods fall short of accurately detecting diverse targets in road maintenance scenarios.

Existing methods often face a trade-off between detection accuracy and computational efficiency, limiting their applicability in real-time UAV-based scenarios. To address these challenges, this study proposes DAL-YOLO, a multi-object detection model based on an improved YOLO11 algorithm, tailored for UAV-based road maintenance tasks. DAL-YOLO achieves a remarkable balance between accuracy and efficiency, with an inference speed of 82 FPS and a computational cost of only 7.4 GFLOPS, ensuring its suitability for practical applications. The key contributions of this study are as follows:

(1) To address the issue of small objects being overlooked in complex backgrounds, the feature extraction module was optimized by integrating Deformable Attention (DAttention) [26] mechanism into the C3k2 module. This enhancement strengthens the representation of key regions and significantly improves small-object detection accuracy.

(2) To overcome insufficient multi-scale feature fusion, an Adaptive Scaled Pyramid model (ASPN) was developed. This improved pyramid structure integrates high-, mid-, and low-level features dynamically, enhancing the detection performance of dense targets in complex scenarios effectively.

(3) To tackle the low recall rates for dense and occluded targets, a lightweight detection head, the Lightweight Shared Conv BatchNorm Head (LSCBNHead), was designed. By combining shared convolutional layers with independent BatchNorm structures, this design improves feature processing efficiency and enhances detection precision and recall. This is the first study to introduce such a detection head for UAV-based multi-object detection in deep learning.

The findings of this study provide an effective solution for addressing multi-object detection challenges in practical road maintenance scenarios. Moreover, this research expands the application potential of UAVs in multi-object detection tasks, offering both theoretical insights and practical value.

2 Research method

2.1 YOLO11

YOLO11 incorporates significant architectural advancements, including the C3k2 and C2PSA components. The C3k2 module represents an evolution of the C2f module introduced in YOLOv8, introducing an adjustable parameter named C3k. When this parameter is set to False, the module retains its original bottleneck design, functioning similarly to the C2f structure. However, when activated (i.e., set to True), the C3k2 module transitions to a C3 structure, thereby enhancing the model's efficiency in processing features. Moreover, the C2PSA component improves the C2f module by integrating a self-attention mechanism (Self-Attention) [2]. This mechanism enables the model to better capture spatial information within input features, thereby improving its ability to understand spatial relationships in the data.

2.2 DAL-YOLO

This study adopts YOLO11 as the base model and introduces three innovative modules-C3k2_DAttention, ASPN, and LSCBNHead-to design an improved detection model, named DAL-YOLO. The overall architecture of DAL-YOLO, as shown in Fig. 1, consists of three main components: the backbone model, the feature pyramid, and the detection head.

Backbone model: To address the challenges of poor detection performance for small objects and feature extraction in complex scenarios, this study integrates an improved deformable attention mechanism into the C3k2 module. By dynamically focusing on key target regions, this

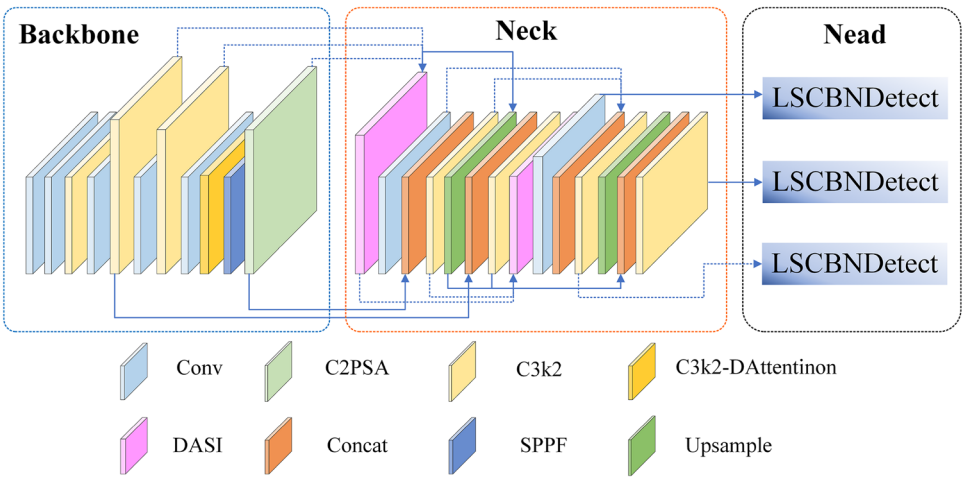


Fig. 1 DAL-YOLO framework diagram. The figure illustrates the architecture of the DAL-YOLO model, including the Backbone, Neck, and Head components

mechanism effectively captures the details of small and low-resolution objects. Unlike traditional attention mechanisms, the proposed method jointly optimizes both the spatial and channel dimensions of feature maps, thereby enhancing feature discriminability. The backbone model comprises four stages, each designed with a lightweight architecture. Furthermore, a multi-head attention mechanism is incorporated into the higher-order stages to further improve adaptability to small objects and complex backgrounds while maintaining a balance between computational efficiency and feature representation.

To improve multi-scale feature fusion, we propose an Adaptive Scaled Pyramid Network (ASPN) that extends beyond traditional FPN and PANet architectures. The core innovation of ASPN is the Dynamic Adaptive Semantic Integration (DASI) [27] module, which addresses the limitations of fixed feature fusion weights in previous methods. The DASI module employs an attention-based adaptive optimization strategy to dynamically adjust the fusion weights between high-level semantic features and low-level spatial features, effectively accommodating targets of varying scales. Unlike conventional static weighting approaches, the DASI module introduces a scale-dynamic mapping matrix over feature maps, enabling adaptive adjustments through efficient parameter learning. When processing UAV-captured complex scenes with dense targets and significant scale variations, the DASI module demonstrates superior performance by enhancing both fine-grained representations of small objects and global context modeling of large objects, resulting in substantial improvements in detection accuracy.

Detection Head: To overcome the bottlenecks in detecting dense and occluded targets, a lightweight detection head, LSCBNHead, is developed. Dense targets often cause feature overlap, while occluded targets result in blurred boundaries and classification difficulties. To address these issues, LSCBNHead reduces computational redundancy by introducing shared convolutional layers. Moreover, independent BatchNorm layers are employed for each detection branch to avoid interference from shared features. By combining multi-scale semantic and spatial features from the backbone model and ASPN, LSCBNHead dynamically adjusts feature weights, enabling more accurate target localization and classification. This design effectively resolves key challenges in detecting dense and occluded targets.

2.2.1 C3k2_DAttention module

The Deformable Attention (DAttention) mechanism enhances feature selection in neural models by dynamically adjusting its receptive field. Unlike traditional attention mechanisms, which compute attention weights at fixed positions, DAttention learns positional offsets to adaptively focus on important features. This flexibility allows the model to better capture critical features in complex scenarios. The structure of DAttention is shown in Fig. 2. The formula for single-scale deformable attention is:

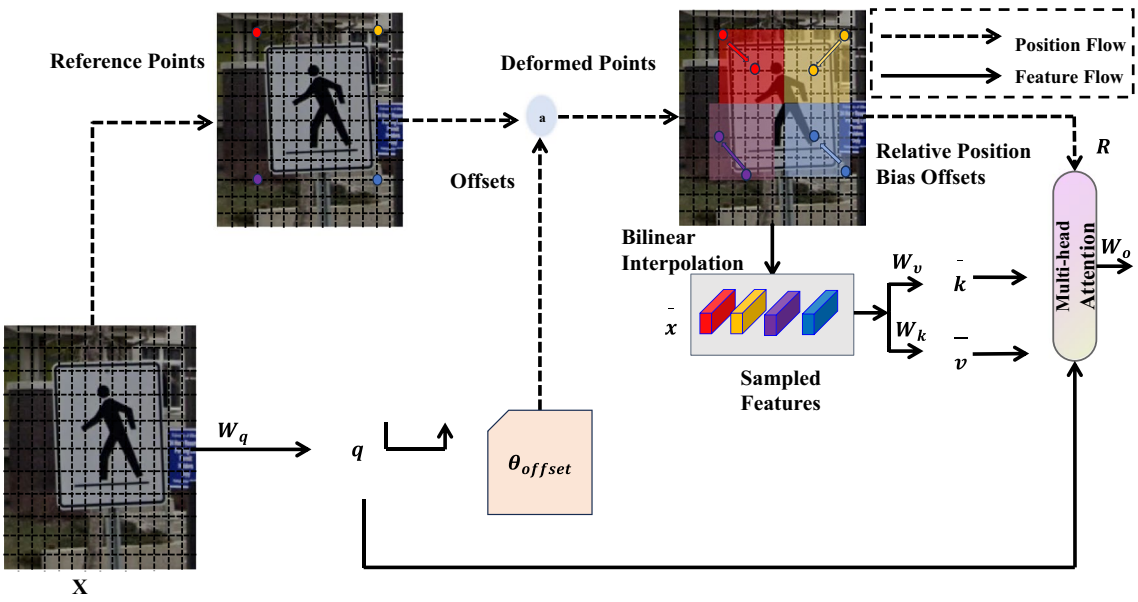


Fig. 2 Deformable attention mechanism structure. The figure shows how DAttention adjusts its focus with learned offsets for better feature extraction

$$DeformAttention(z_q, p_q, x) = \sum_{m=1}^{M} W_m \left[\sum_{k=1}^{K} A_{mqk} \cdot W'_m \cdot x\left(p_q + \Delta p_{mqk}\right) \right], \quad (1)$$

where: P_q is the reference position of the input feature Z_q, represented as a 2D vector. ΔP_{mqk} denotes the positional offset of the K-th sampling point relative to the reference position in the m-th attention head. (A_{mqk}) are the learned attention weights, computed through a fully connected layer, avoiding the traditional Query-Key dot product.

This mechanism dynamically aggregates features from sampled positions $(p_q + \Delta p_{mqk})$, with learned offsets, and uses attention coefficients (A_{mqk}) to focus on relevant regions of the input data.

The DAttention mechanism is integrated into the C3k2 module to create the C3k2_DAttention module, as shown in Fig. 3. By placing DAttention within the bottleneck of the C3k2 module, we enable adaptive focusing on critical spatial features. To maintain computational efficiency, the C3k2_DAttention module is applied selectively in the eighth layer of the backbone, where high-level global context is most beneficial. This selective integration improves small-object detection and feature discrimination without incurring significant computational overhead.

In summary, integrating DAttention into the C3k2 module improves the model's ability to focus on key regions, enhancing detection accuracy in complex scenarios. This integration strikes a balance between computational efficiency and detection performance, making it suitable for real-time UAV-based applications.

2.2.2 ASPN

Traditional feature pyramids face challenges such as inefficient feature transmission, limited contextual information capture, and unbalanced multi-scale feature fusion. These issues arise from their reliance on single connection methods, which result in inadequate interaction between high-level semantic features and low-level spatial details. As a result, performance in complex scenarios, particularly for small and multi-scale object detection, remains suboptimal [20].

To address these limitations, we propose the Adaptive Scaled Pyramid (ASPN) model, which integrates the Dynamic Adaptive Semantic Integration (DASI) module [27]. The DASI module uses a dimension-aware selective integration mechanism to enable dynamic and precise fusion of multi-scale features. Additionally, ASPN incorporates a feature diffusion mechanism that enhances the interaction and transmission of information between high-level and

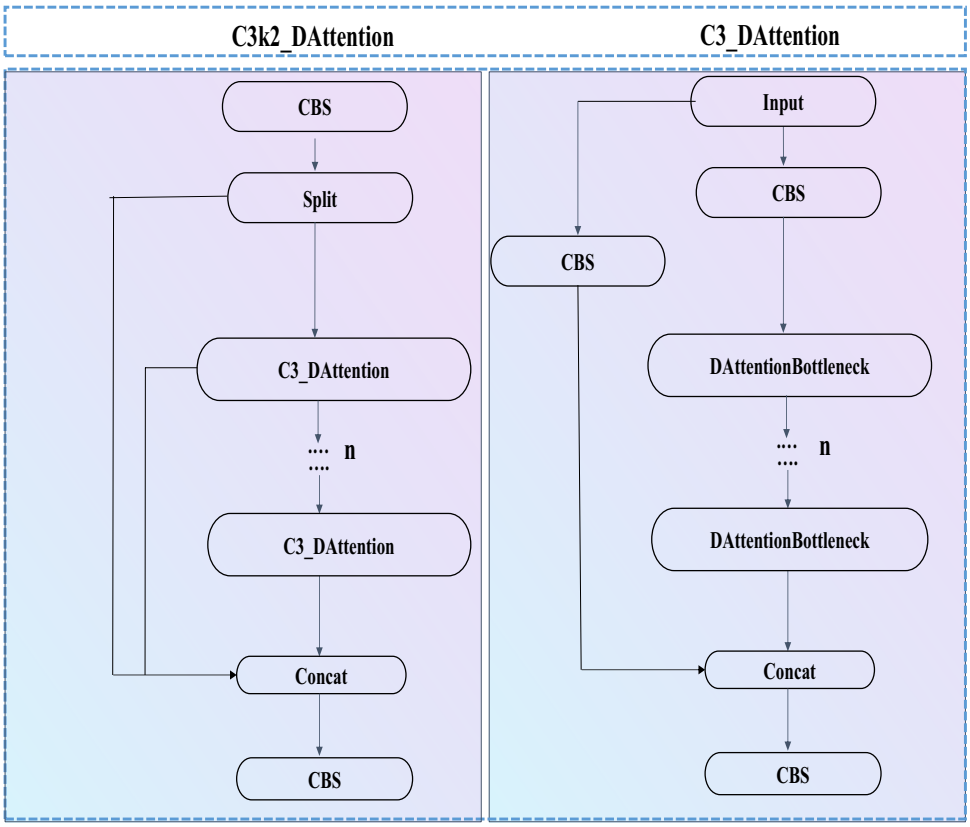


Fig. 3 C3k2_DAttention module structure. The figure shows the integration of DAttention with C3k2 for enhanced detection performance

low-level features. This mechanism allows feature information to propagate across scales, with high-level semantic features flowing downward and low-level spatial details being fed back upward, ensuring balanced feature fusion and context preservation. The integration of residual connections and feature weighting strategies further improves information retention and efficient multi-scale feature interaction. The structure of ASPN, including the DASI module, is shown in Fig. 4.

Feature Extraction and Fusion: The ASPN model employs a parallel convolutional kernel design, utilizing kernels of different sizes to capture both local details and global context. This design enhances the richness of feature representations. The features are then fused and downsampled, with residual connections applied to preserve the integrity of the features. These connections add input features element-wise to the convolved features, ensuring complete information transmission. Additionally, a feature weighting strategy dynamically assigns weights to output features based on their importance, further enhancing the multi-scale feature diffusion process.

DASI Module: The DASI module addresses challenges in detecting small objects, especially in complex scenarios where objects may be densely packed or obscured. During multi-level downsampling in object detection, small-object details are often lost in high-dimensional features, while low-dimensional features lack sufficient context. To overcome this, the DASI module adaptively selects and combines features from multiple dimensions, preserving and enhancing small-object information. The structure of the DASI module is depicted in Fig. 5.

The DASI module performs feature fusion as follows: first, the high-dimensional feature F_h and low-dimensional feature F_l are aligned with the current layer feature F_u through convolution and interpolation operations.

The features are then divided into three segments along the channel dimension: h_i, l_i and u_i.

Each segment undergoes fusion using an adaptive weight selection mechanism. The fusion is expressed as:

$$\hat{F}_u = \delta\left(\beta\left(\mathrm{Conv}\left([\alpha u_i + (1-\alpha)h_i]_{i=1}^{4}\right)\right)\right). \tag{2}$$

Here, α represents the weight values obtained by applying the sigmoid activation function to u_i, and h_i denotes the input feature map. The convolution operation Conv() is applied to the linear combination of αu_i and h_i^4, where h_i^4 refers to the feature map raised to the fourth power. The parameters β and δ correspond to batch normalization (BN) and the non-linear activation function, respectively.

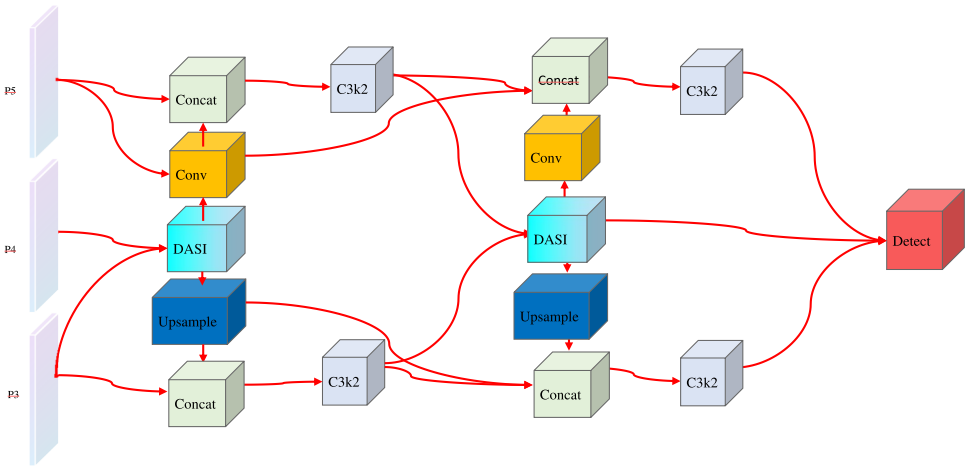


Fig. 4 ASPN structure diagram. The figure illustrates the structure of the ASPN, incorporating the DASI module for multi-scale feature fusion and enhanced contextual information capture

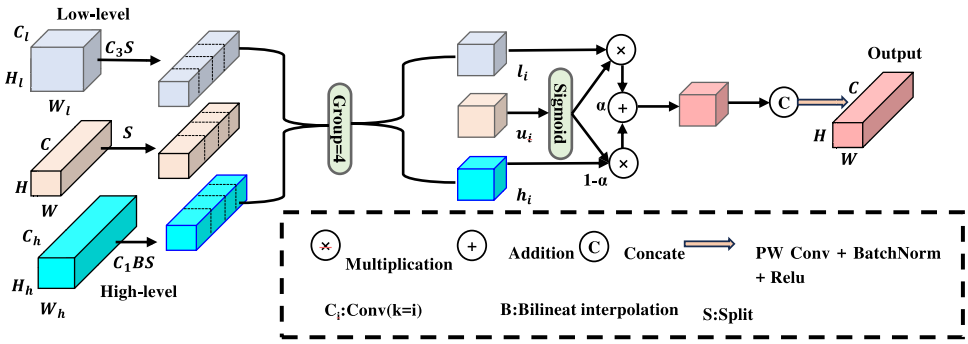


Fig. 5 DASI structure diagram. The figure shows the structure of the dynamic DASI module, which adaptively combines high-dimensional and low-dimensional features for improved small-object detection

This fusion mechanism enables the DASI module to effectively combine high-dimensional and low-dimensional features, improving the detection of small objects, particularly in complex scenarios.

2.2.3 LSCBNHead

The original YOLO11 detection head has some inherent limitations. First, it contains a large number of parameters, which significantly increase the computational cost of the model. Second, although it utilizes a feature pyramid for multi-scale feature fusion, it does not fully exploit the multi-scale features, leading to suboptimal detection performance, especially for small and large objects. To address these issues, we propose a novel detection head structure, called LSCBNHead. The core design and optimization of LSCBNHead are outlined below and illustrated in Fig. 6:

(1) In the original detection head, independent convolution layers are used for each scale (P3, P4, P5), which can lead to redundancy and increased computational cost. In LSCBNHead, these independent convolution layers are replaced by shared convolution layers. First, a 1*times*1 convolution is applied to reduce the feature map channels, significantly lowering computational overhead. Then, a shared 3*times*3 convolution is employed to extract multi-scale features from all scales (P3, P4, P5) simultaneously. Sharing convolution weights across scales reduces the number of parameters and redundant computations, while ensuring consistent feature extraction. This design enhances the model's generalization capability and improves its performance across different object sizes.

(2) After the shared convolution, we introduce independent BatchNorm layers for each detection branch corresponding to the different scales (P3, P4, P5). These independent BatchNorm layers allow each detection branch to normalize its features separately, ensuring that different scales do not interfere with each other. This design helps maintain optimal feature distributions for each scale, improving both classification and regression performance for small and large objects.

The optimizations in LSCBNHead reduce the number of parameters and computational complexity while significantly enhancing the model's ability to perceive multi-scale features. The use of shared convolutions reduces redundancy, and the independent BatchNorm layers ensure that each scale maintains distinct feature distributions. As a result, LSCBNHead improves detection performance for both small and large objects, especially in complex scenarios.

3 Experiment and analysis

3.1 Dataset

The experimental data for this study were obtained from the UAVROAD dataset [31], developed by Southeast University for UAV-based daily road maintenance tasks. To ensure the fairness of the experiments, the dataset was divided into training, validation, and testing sets in a ratio of 8:1:1. Specifically, the training set includes 4001 images, while the validation and testing sets contain 500 and 501 images, respectively. All experimental results reported in this paper are based on the testing set. Example images from each category are shown in Fig. 8, and the distribution of the number of images in each category is depicted in Fig. 7.

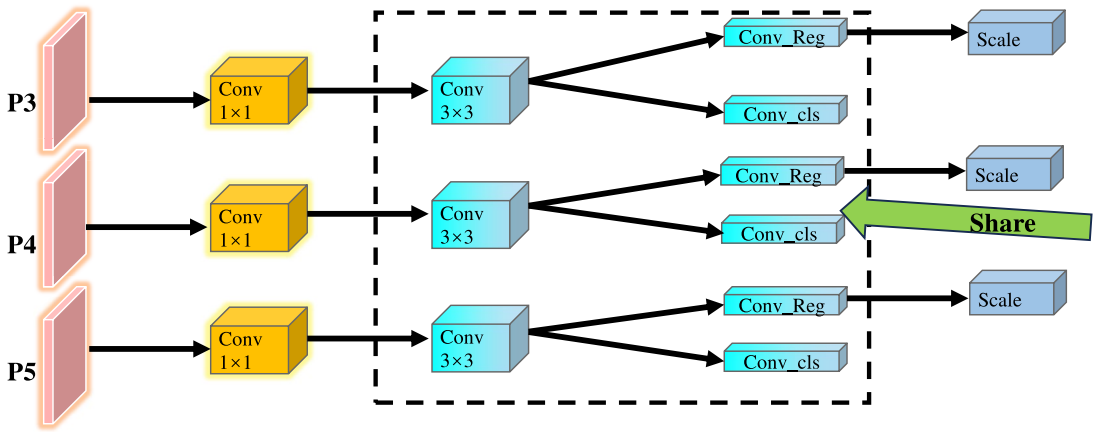


Fig. 6 LSCBNHead structure diagram. The figure shows the shared convolution design and independent BatchNorm layers for each scale, improving multi-scale detection performance

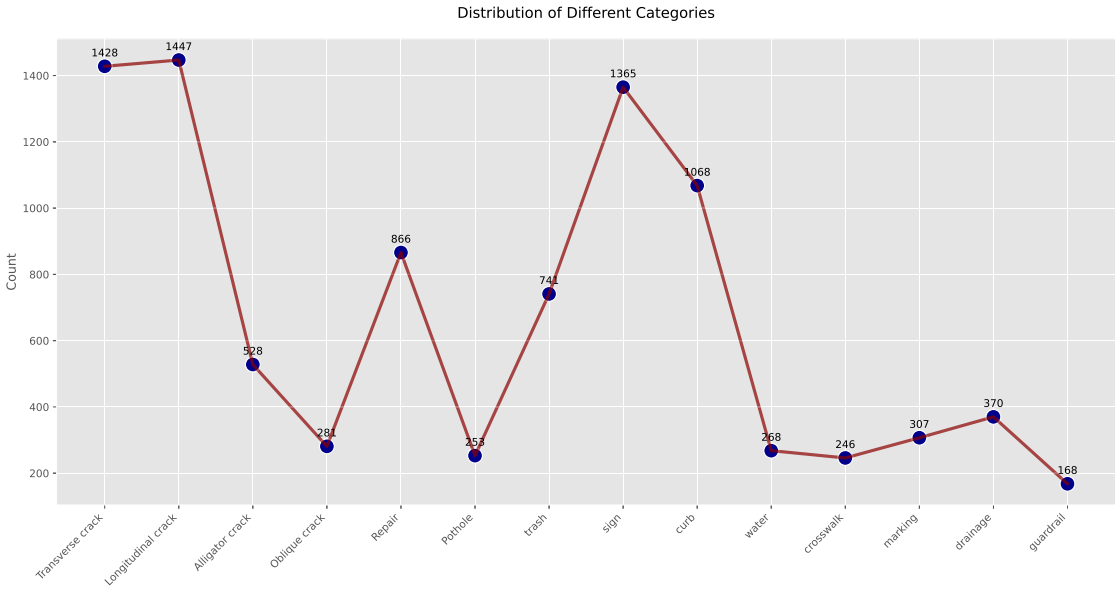


Fig. 7 Distribution of image categories in the dataset

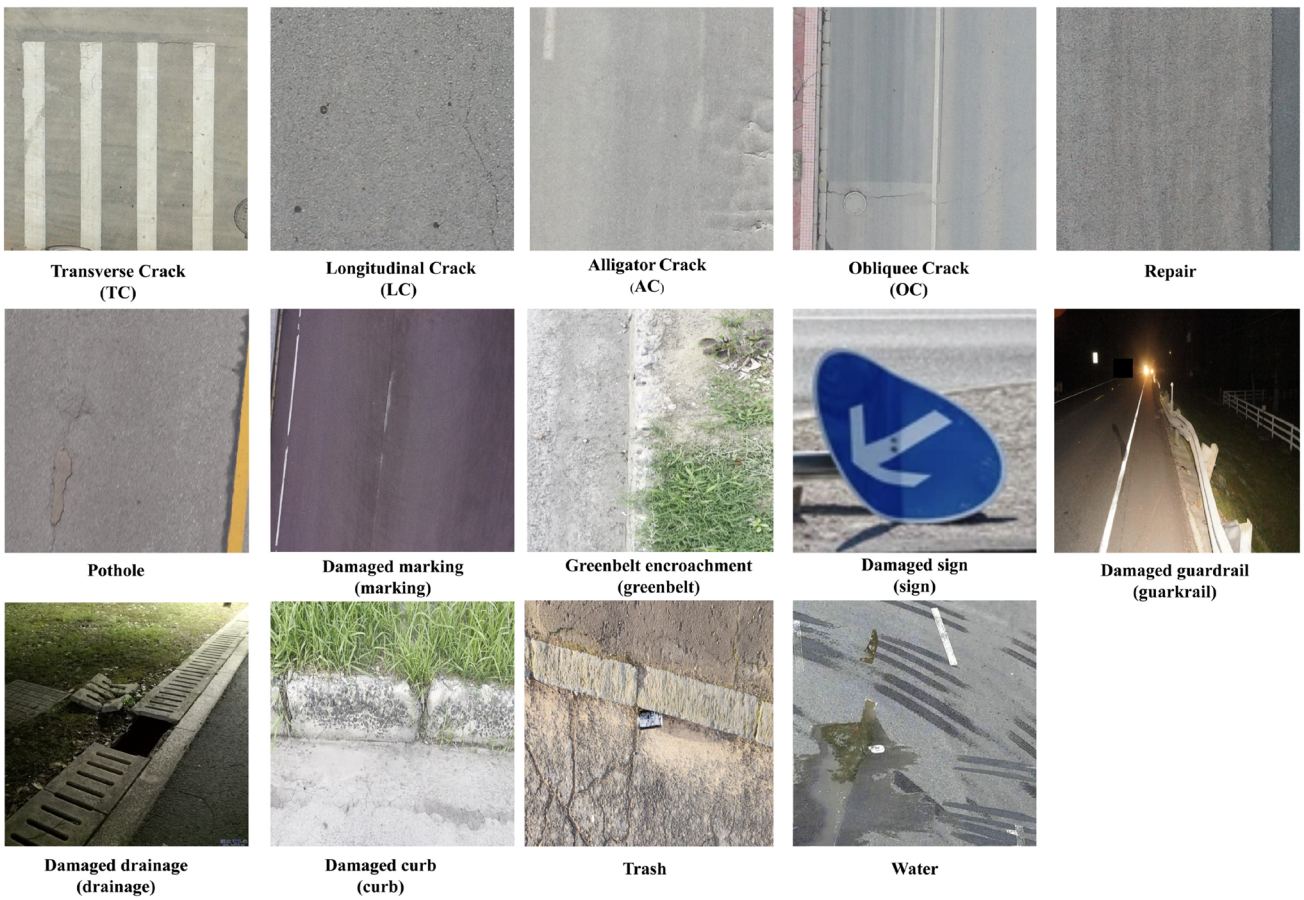


Fig. 8 Example images from each category in the dataset

3.2 Evaluation indicators

To accurately evaluate the performance of the proposed model in UAV-based routine road maintenance environments, this study uses mean Average Precision (mAP), Recall, Precision, and Localization Recall Precision (LRP) as the primary evaluation metrics.

Precision measures the proportion of correctly predicted positive samples among all detected samples:

$$Precision = \frac{TP}{TP + FP}. \tag{3}$$

Average Precision (AP) is the average of Precision across all images:

$$AP = \frac{\sum Precision}{images}, \tag{4}$$

mean Average Precision (mAP) is the mean of AP values for all classes:

$$mAP = \frac{1}{c}\sum_{i=1}^{c} AP_i. \tag{5}$$

Localization Recall Precision (LRP) evaluates the model's localization performance by considering both the accuracy and recall of the predicted bounding boxes:

$$\mathrm{LRP} = \frac{1}{N_{\mathrm{TP}} + N_{\mathrm{FP}} + N_{\mathrm{FN}}}\left(\sum_{i=1}^{N_{\mathrm{TP}}} \varepsilon_{\mathrm{Loc}}(i) + N_{\mathrm{FP}} + N_{\mathrm{FN}}\right), \tag{6}$$

where $\varepsilon_{\mathrm{Loc}}(i)$ is the localization accuracy of the $i-$th True Positive (TP) detection.

Additionally, model efficiency is evaluated using inference speed (FPS) and computational cost (GFLOPs) to assess effectiveness and feasibility.

3.3 Experimental environment and parameter settings

Table 1 presents the experimental environment configurations, while Table 2 provides the hyperparameter settings used in the experiments.

Table 1 Experimental environment configuration

Configuration	Parameter
Operating system	Ubuntu 18.04
GPU	RTX4090 (24 G)
Programming language	Python 3.10.14
Development framework	PyTorch 2.2.2
Platform	CUDA 12.1

Table 2 Hyperparameter settings

Parameter name	Parameter value
Learning rate	0.01
Batch size	16
Image size	640 × 640
Epochs	300
Momentum	0.937
Optimizer	SGD

3.4 Ablation experiment

To evaluate the effectiveness of the improved modules integrated into DAL-YOLO, a series of ablation experiments were conducted on the UAVROAD dataset. Table. 3 compares the performance of the original model with the modified versions incorporating the proposed modules: A represents ASPN, B represents LSCBNHead, and C represents C3k2_DAttention.

The original YOLO11 model achieved an AP of 34.5%. After incorporating the ASPN module, the AP increased to 37.4%, with small object detection performance improving significantly from 9.1 to 12.7%. This indicates that the ASPN module effectively enhances multi-scale feature fusion through its adaptive spatial pyramid model, particularly benefiting small object detection. Upon further integrating the LSCBNHead module, the AP improved to 38.6%, with AP_S reaching 13.5%. Furthermore, the parameter count decreased from 2.68 to 2.56 M, and computational complexity was optimized, demonstrating that the shared convolution and independent BatchNorm design effectively enhance detection performance while reducing model complexity. Finally, incorporating the C3k2_DAttention module resulted in an AP of 39.8%, with small object detection performance improving to 14.6%, and medium- and large-object detection accuracies increasing to 38.7% and 58.8%, respectively. This highlights the advantage of deformable attention mechanisms in enhancing feature extraction and multi-scale perception. Overall, the ASPN, LSCBNHead, and C3k2_DAttention modules not only improve detection

Table 3 Ablation experiment

Methods	AP (%)	AP_S (%)	AP_M (%)	AP_L (%)	GFLOPs
YOLO11n	34.5	9.1	33.6	54.2	6.3
A	37.4	10.2	36.8	57.8	7.4
B	35.8	9.1	35.7	54.6	6.4
C	35.5	11.6	33.5	55.1	6.4
A+B	38.6	11.5	37.5	56.2	7.3
A+B+C	39.8	14.6	38.7	58.8	7.4

accuracy but also maintain low parameter counts, demonstrating their effectiveness and practicality in multi-object detection tasks.

To further validate the effectiveness of attention mechanisms in object detection tasks, a comparative study was conducted on different attention mechanisms to analyze their performance in feature extraction and multi-scale perception. As demonstrated earlier, the introduction of attention mechanisms significantly improves detection accuracy, particularly for small objects. This comparison explores the strengths and limitations of various attention mechanisms, evaluating their potential for enhancing model performance, optimizing feature representation, and reducing computational complexity. The experimental results are shown in Table. 4.

Table 4 Comparison of different feature extraction mechanisms integrated into the C3k2 module

Method	AP (%)	AP50 (%)	FPS	GFLOPs
YOLO11n	34.5	54.3	136	6.3
C3k2-EMA	35.2	53.5	114	6.8
C3k2-SE	33.8	54.6	132	6.3
C3k2-ECA	32.9	53.9	119	6.4
C3k2-MLCA	33.2	53.2	121	6.5
C3k2-DAttention	35.8	55.2	124	6.4

In the previous sections, we compared and analyzed the overall performance of different attention mechanisms using quantitative metrics such as AP, AP50, and FPS. However, global performance metrics alone cannot fully capture the role of attention mechanisms in the feature extraction process, particularly their ability to focus on target regions. To further investigate the specific impact of different attention mechanisms on feature extraction and target perception, heatmap comparisons were generated, as shown in Fig. 9. Higher activation values indicate a greater probability of a target being present in a specific location, which is represented as brighter and more prominent areas in the heatmaps. Specifically, subfigure (a) corresponds to C3k2_DAttention, (b) to the YOLO11 baseline, (c) to C3k2_EMA [14], (d) to C3k2_SE, (e) to C3k2_ECA [25], and (f) to C3k2_MLCA [22].

As shown in the figure, the C3k2_DAttention module produces feature maps with heatmaps that focus more on relevant target regions, highlighting its effectiveness in feature extraction for the dataset.

After completing the comparison of attention mechanisms, we further investigated the impact of feature pyramid modules on detection performance. As a critical component of object detection models, the design of the feature pyramid structure directly influences the ability to represent multi-scale features. Using the same baseline model (YOLO11) and experimental settings, we compared six structures: Baseline, BiFPN [19], SlimNeck [11], GFPN

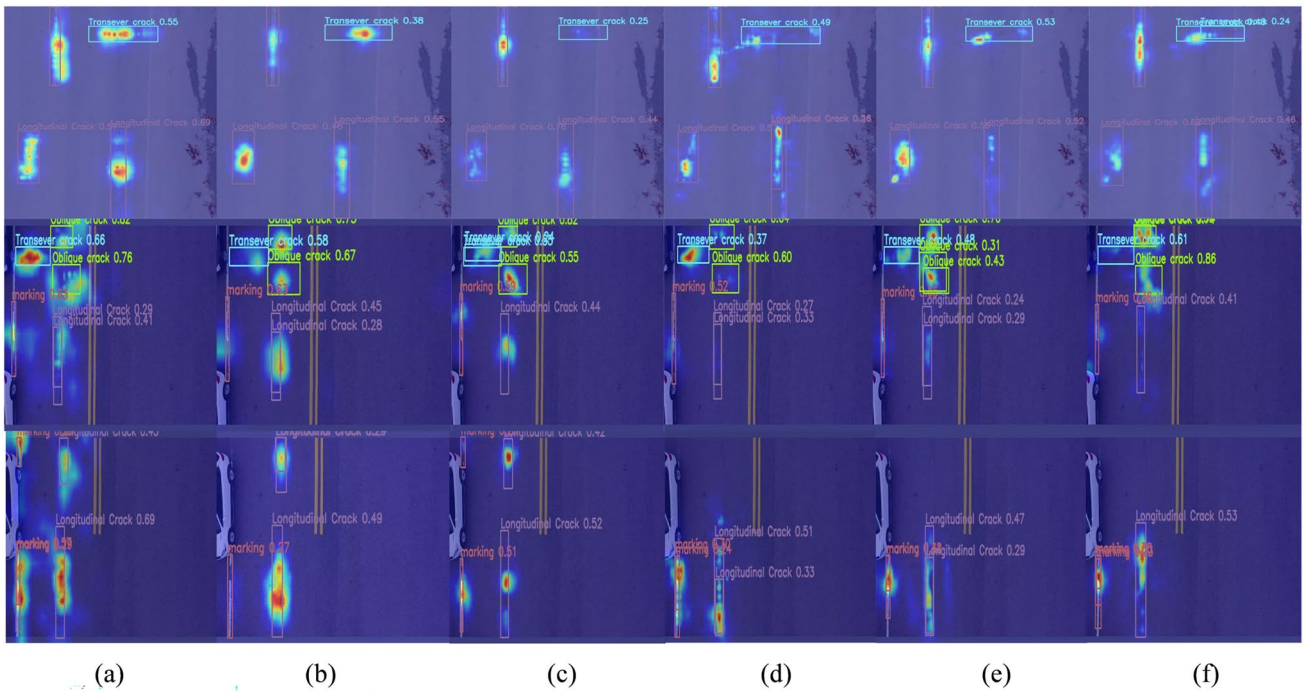


Fig. 9 Comparison of heatmaps with different attention mechanisms. Subfigures (**a**) to (**f**) show the focus of various attention modules on target regions

[7], EfficientRepBiPAN [10], and ASPN. The evaluation systematically assessed their detection performance (AP and AP50), GFLOPS, and FPS, as summarized in Table 5.

In the comparison of feature pyramid modules (Table 5), the ASPN module demonstrated the best performance, with AP and AP50 improved by 2.9% compared to the baseline. This highlights its significant advantage in multi-scale feature fusion capability. Moreover, ASPN maintained a favorable balance between GFLOPS (7.4) and FPS (104), showcasing excellent overall performance. The BiFPN module also achieved noticeable improvements in AP and AP50, demonstrating the effectiveness of weighted multi-scale feature processing, although its inference speed was slightly lower than the baseline. SlimNeck, with the lowest GFLOPS (6.1) and relatively high FPS (131), achieved a lightweight design; however, its detection accuracy fell below the baseline, limiting its applicability in tasks requiring high precision. GFPN and EfficientRepBiPAN exhibited moderate performance improvements, but their high GFLOPS values resulted in slower inference speeds, making them more suitable for scenarios prioritizing accuracy over real-time requirements. Overall, the ASPN module achieved the optimal balance between performance and efficiency, making it the preferred choice for multi-object detection tasks.

Table 5 Comparison of feature pyramid modules

Methods	AP (%)	AP50 (%)	FPS	GFLOPS
YOLO11n	34.5	54.3	136	6.3
BiFPN	35.6	55.2	122	6.4
SlimNeck	33.2	52.6	131	6.1
GFPN	34.8	54.9	94	8.5
EfficientRepBiPAN	34.1	53.2	98	8.3
ASPN	37.4	57.2	104	7.4

3.5 Comparative experiment

The comparative results of different models on the UAVROAD dataset are shown in Table 6. All comparative experiments were conducted under a unified experimental environment to ensure fairness. The specific experimental parameters, including optimizer settings and model input sizes, are detailed in Tables 1 and 2. The Input Size for each model is presented in Table 6. All models were trained with the SGD optimizer, except for RT-DETR, which uses the AdamW optimizer. This ensures a fair comparison of the models' performance. DAL-YOLO achieves state-of-the-art performance with 39.8% AP, 58.8% AP50, and 46.7% LRP, while maintaining competitive efficiency at 82 FPS with only 7.4 GFLOPS.

Among the latest state-of-the-art methods, DAL-YOLO demonstrates superior or competitive performance across various metrics. Compared to the EfficientDet series [16], DAL-YOLO shows comparable accuracy to EfficientDet-D2 (39.8% vs. 39.5% AP), while achieving significantly higher inference speed (82 FPS vs. 54 FPS). Moreover, DAL-YOLO outperforms the transformer-based detector RT-DETR [29] by 3.9 percentage points in AP (39.8% vs. 35.9%) and 3.9 percentage points in AP50 (58.8% vs. 54.9%), while being 1.7*times* faster in inference speed with substantially lower computational cost (7.4 GFLOPS vs. 102.2 GFLOPS). In comparison to other advanced architectures like Mamba-YOLO-B [24] and Hyper-YOLO [4],

Table 6 Comparison of different models on the UAVROAD dataset

Model	Input size	LRP (%)	AP (%)	AP50 (%)	AP_S (%)	AP_M (%)	AP_L (%)	FPS	GFLOPS
YOLO11n	640 × 640	41.1	34.5	54.3	9.1	33.6	54.2	136	6.3
UM-YOLO	640 × 640	40.7	32.8	51.8	9.3	34.0	53.8	70	7.7
[5]	640 × 640	39.2	31.9	51.5	7.9	29.7	48.2	98	15.52
RetinaNet	800 × 1333	37.5	33.2	53.4	7.8	30.1	48.8	46	45.2
Faster R-CNN	800 × 1333	37.3	32.4	52.1	7.5	28.2	46.4	21	60.2
RT-DETR	640 × 640	42.6	35.9	54.9	11.9	33.2	55.8	48	102.2
EfficientDet-D0	512 × 512	36.7	30.2	48.7	5.4	28.8	47.8	82	2.5
EfficientDet-D1	640 × 640	38.5	34.7	52.1	9.5	33.9	52.6	68	3.4
EfficientDet-D2	768 × 768	41.2	39.5	58.1	14.2	38.1	57.4	54	7.3
YOLOv8n	640 × 640	39.7	32.8	52.7	8.4	31.9	50.9	128	8.2
YOLOv5 s	640 × 640	40.4	33.1	51.1	8.2	30.8	49.6	102	15.8
[28]	640 × 640	36.7	33.1	53.7	8.6	32.3	51.4	45	49.2
Mamba-YOLO-B	640 × 640	46.3	38.2	57.4	13.6	37.1	56.9	61	44.5
Hyper-YOLO	640 × 640	45.2	37.8	56.8	12.5	36.4	55.8	122	9.5
[23]	640 × 640	37.5	31.5	37.8	8.1	30.5	49.2	108	13.1
DAL-YOLO	640 × 640	46.7	39.8	58.8	14.6	38.7	58.8	82	7.4

DAL-YOLO shows consistent improvements in AP, achieving 1.6 and 2.0 percentage points higher AP respectively, while maintaining or exceeding their efficiency.

For small object detection, which is crucial in UAV-based applications, DAL-YOLO achieves 14.6% AP_S, surpassing EfficientDet-D2 (14.2%), Mamba-YOLO-B (13.6%), RT-DETR (11.9%), and Hyper-YOLO (12.5%). Similar advantages are evident in medium and large object detection, where DAL-YOLO reaches 38.7% AP_M and 58.8% AP_L, demonstrating the effectiveness of its adaptive feature fusion strategy.

When compared to traditional models like Faster R-CNN (32.4% AP, 21 FPS) and RetinaNet (33.2% AP, 46 FPS), as well as lightweight models like YOLOv5 s [21] (33.1% AP, 102 FPS) and YOLOv8n [8] (32.8% AP, 128 FPS), DAL-YOLO shows substantial improvements in detection accuracy while maintaining competitive inference speed.

To further illustrate the performance trade-offs, Fig. 10 presents a speed-accuracy comparison. The left panel shows DAL-YOLO's optimal balance between accuracy (AP) and speed (FPS), with bubble sizes representing GFLOPS to highlight its efficient computational cost. The right panel confirms DAL-YOLO's consistent advantage across different precision metrics, particularly in scenarios requiring higher localization accuracy.

3.6 Model generalization experiment

To evaluate the generalization ability of the proposed model, DAL-YOLO, this experiment uses the VisDrone2019 [30] dataset for testing. This dataset contains drone-captured images from various complex environments, covering different types of objects and diverse scene conditions, which allows for an effective assessment of the model's performance in various challenging scenarios. By validating on the VisDrone 2019 dataset, we can comprehensively understand the generalization ability of DAL-YOLO, ensuring its stability and efficiency in real-world applications. The experimental results on the VisDrone 2019 dataset are summarized in Table 7, providing detailed comparisons and demonstrating the model's effectiveness across different scenarios. The Hyperparameter settings used in this experiment can be found in Tables 1 and 2.

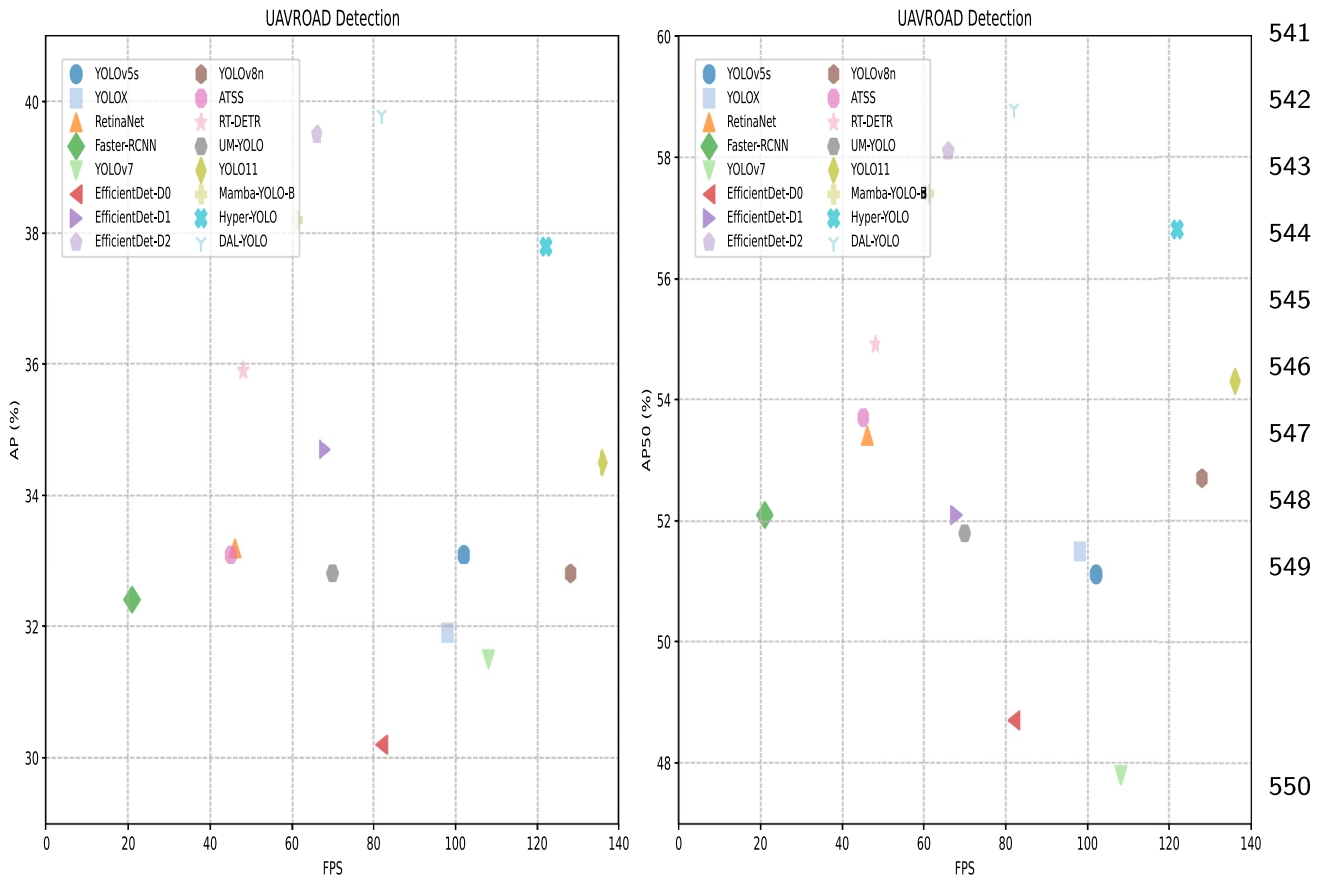


Fig. 10 Visualization of comparative experiments. A comparison of different models in terms of accuracy and inference time across various conditions

Table 7 Comparison of different models on the VisDrone dataset

Model	AP50 (%)	AP75 (%)	LRP (%)	FPS	GFLOPS
YOLO11n	40.1	24.4	37.2	122	6.4
EfficientDet-D0	32.6	21.4	34.7	76	2.7
RT-DETR	41.5	25.2	38.1	36	103.5
Mamba-YOLO-B	40.7	24.9	37.4	52	44.5
Faster R-CNN	33.5	18.4	29.7	15	62.9
DAL-YOLO	41.9	25.7	38.6	71	7.4

In the generalization experiment, DAL-YOLO demonstrates its strong adaptability across different datasets. Compared to other models, DAL-YOLO achieves the highest AP50 (41.9%) and AP75 (25.7%) on the UAVROAD dataset, showing its ability to detect objects accurately in a variety of conditions. Furthermore, DAL-YOLO excels in LRP (38.6%), indicating its robust performance in localizing objects, which is critical in dynamic and complex environments. The model also maintains competitive FPS (71) and GFLOPS (7.4), ensuring that high performance is achieved without compromising efficiency. This strong generalization ability makes DAL-YOLO highly effective for a wide range of UAV-based tasks, ensuring reliable performance in diverse scenarios.

4 Visualization

To further validate the effectiveness of DAL-YOLO, a visual analysis was conducted to compare the performance of different models across various scenarios, as illustrated in Fig. 11.

As shown in Fig. 11, DAL-YOLO significantly outperforms other models in terms of both detection accuracy and missed detection rates across various challenging scenarios:

Small and Low-Contrast Targets: DAL-YOLO demonstrates superior performance in detecting small and low-contrast objects, such as longitudinal cracks, transverse cracks, and water markings. These are critical for road maintenance but often overlooked by YOLO11

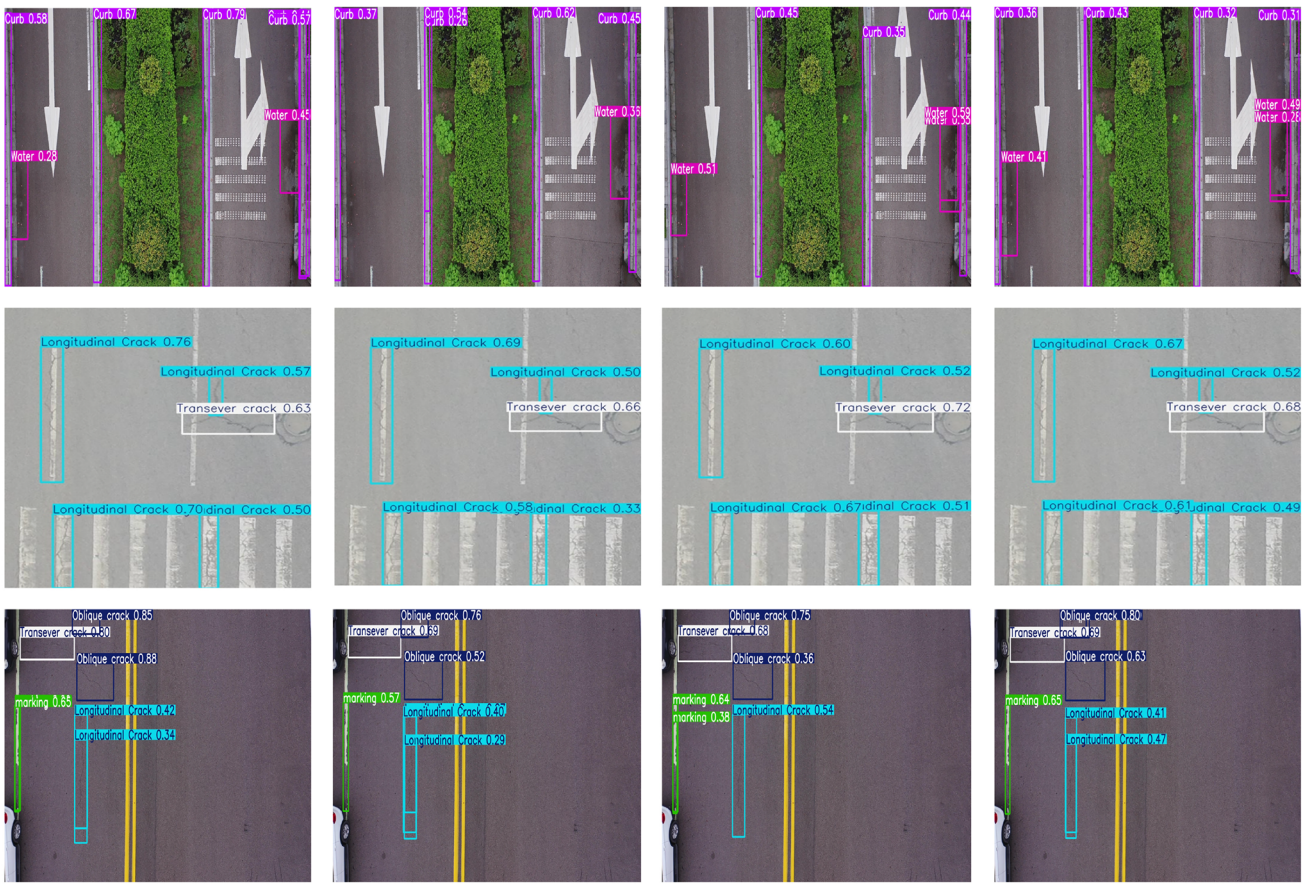


Fig. 11 Visual comparison of DAL-YOLO with YOLO11, Faster R-CNN, and RT-DETR under various road maintenance scenarios

and Faster R-CNN, as evidenced by missed detections and lower confidence scores in these models. For example, DAL-YOLO identifies a longitudinal crack with a confidence score of 0.76, while YOLO11 and Faster R-CNN yield lower scores (0.60 and 0.52, respectively), and RT-DETR partially misses the target.

Occlusion and Dense Targets: DAL-YOLO excels in scenarios with occluded or overlapping objects, such as when detecting cracks near road markings or dense objects along curbs. In Fig. 11, DAL-YOLO successfully identifies multiple cracks and curbs with high confidence, whereas YOLO11 and Faster R-CNN either fail to detect these objects or provide significantly lower confidence levels. In the second row of Fig. 11, DAL-YOLO detects both transverse cracks and longitudinal cracks simultaneously with confidence scores above 0.7, while RT-DETR struggles to distinguish between them.

Complex Backgrounds: The ability to accurately detect objects against complex backgrounds is another key strength of DAL-YOLO. In the bottom row of Fig. 11, where road markings and cracks are superimposed on a textured road surface, DAL-YOLO maintains high precision, detecting both oblique cracks and longitudinal cracks with confidence scores above 0.85, whereas YOLO11 and Faster R-CNN show incomplete or false detections.

Quantitative Alignment with Visual Results: These visual findings align well with the quantitative results presented earlier. DAL-YOLO achieves an AP of 39.8% and significantly outperforms YOLO11 and Faster R-CNN in AP_S, indicating its robustness in small-object detection. This visual analysis further substantiates DAL-YOLO's ability to achieve high detection accuracy under real-world UAV scenarios.

The visual evidence not only highlights the robustness and adaptability of DAL-YOLO but also demonstrates its suitability for real-time applications, such as road maintenance inspections, where detecting diverse and complex targets is essential.

However, despite DAL-YOLO's strong performance across various road maintenance scenarios, its performance deteriorates under the interference of noise, leading to missed detections. As shown in Fig. 12, the red boxes highlight the missed detections, where the model fails to detect certain critical objects, such as small trash near the drain, due to the surrounding clutter and noisy background. These missed detections occur because the noise and overlapping objects reduce the model's ability to accurately recognize



Fig. 12 DAL-YOLO with missed detections (red boxes) due to noise and clutter in road maintenance scenarios

small targets. This failure case demonstrates the challenges DAL-YOLO faces in noisy environments, and future work will focus on improving its performance in handling such scenarios.

5 Embedded platform deployment

To validate the real-time detection performance of DAL-YOLO, this paper deploys the improved model onto an edge device, the NVIDIA Jetson Orin Nano, for real-time detection via camera feed. Its technical specifications are shown in Table 8.

As a compact embedded artificial intelligence computing device, the Jetson Orin Nano offers a variety of connectivity options, including camera interfaces, GPIO pins, USB ports, and more, making it suitable for a wide range of application scenarios. This device can be embedded into systems such as drones, small robots, and smart cameras, making it ideal for the edge deployment tasks of the improved model in this paper. The Jetson Orin Nano device is shown in Fig. 13.

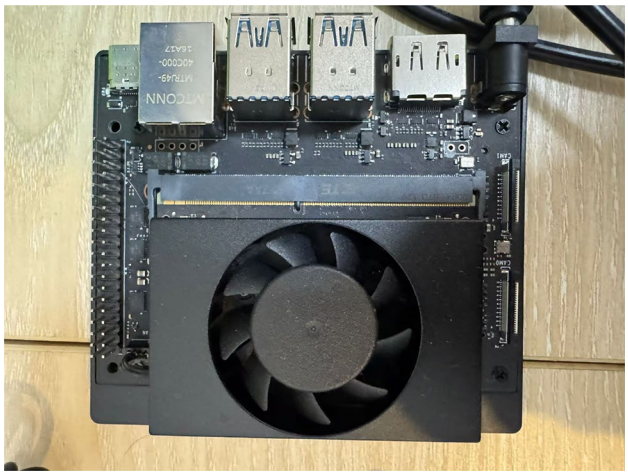


Fig. 13 Jetson Orin Nano equipment

6 Conclusion

This study presents DAL-YOLO, a novel multi-object detection model tailored for UAV-based road maintenance scenarios by integrating Deformable Attention mechanism into the C3k2 module, the proposed method enhances the representation of small objects in complex backgrounds. Additionally, the Adaptive Scaled Pyramid Network (ASPN) dynamically fuses multi-scale features, improving the detection performance for dense and diverse targets. The Lightweight Shared Convolution BatchNorm Head (LSCBNHead) further optimizes feature processing efficiency, enhancing the precision and recall for occluded and densely packed objects.

Experimental results on the UAVROAD dataset demonstrate that DAL-YOLO achieves state-of-the-art performance, with an AP of 39.8% and an AP50 of 58.8%, while maintaining an efficient inference speed of 82 FPS with a computational cost of only 7.4 GFLOPS. The model consistently outperforms existing methods in detecting small, medium, and large objects, highlighting its ability to balance detection accuracy and computational efficiency. These results validate DAL-YOLO's suitability for real-time UAV applications in routine road maintenance tasks.

In addition to improving multi-object detection in challenging UAV scenarios, this research provides a theoretical foundation and practical guidance for applying deep learning models to road maintenance. The proposed framework can be extended to other UAV-based inspection tasks, such as insulator defect detection and environmental surveillance.

Future work will focus on further optimizing the model's performance in scenarios with noise and various degrees of blurring, which currently challenge its detection accuracy. Additionally, efforts will be directed toward enhancing the model's robustness in such environments by exploring noise-robust techniques and improving feature extraction in complex, degraded conditions.

Table 8 Technical specifications of Jetson Orin Nano

Name	Specification
GPU	NVIDIA Volta™ Architecture, 384 NVIDIA CUDA® cores
CPU	6-core NVIDIA Carmel ARM®v8.2 64-bit CPU
Storage	16 GB eMMC 5.1
Memory	8 GB 128-bit LPDDR4x
Network	Gigabit Ethernet, onboard WiFi/BT
Interface	4 USB 3.1 ports, HDMI output, eSATA port, 2 CSI ports, GPIO port
Dimensions	69.6 mm × 45 mm
Power consumption	10 W/15 W

Author Contributions Xuerui Lan designed the study, conducted the experiments, and performed data analysis. Lijun Liu assisted in data collection, literature review, and provided critical feedback during the manuscript preparation. Xuyang Wang supervised the study, contributed to the methodology, and revised the manuscript. Xuerui Lan and Lijun Liu wrote the main manuscript text, and Xuyang Wang prepared figures and tables. All authors reviewed and approved the final manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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